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ORIGINAL ARTICLE

Leptospira spp. in Small Mammals from Areas with Low and High Human Hantavirus Incidences in South-West Germany

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Abstract

Introduction: Leptospirosis is caused by *Leptospira* spp. and is considered the most widespread zoonotic disease worldwide. It mimics nephropathia epidemica in humans, a disease mainly caused by Puumala hantavirus (PUUV). Small mammals are reservoirs for *Leptospira* spp. and PUUV. Seewis virus (SWSV) is a shrew-borne hantavirus with unknown pathogenicity. The objective of this study was to estimate the prevalence for *Leptospira* spp. and the frequency of *Leptospira*-hantavirus co-infections in small mammals collected at locations with high and low incidences in humans.

Materials and Methods: In 2012 and 2013, 736 small mammals belonging to seven species (*Apodemus flavicollis*, *Microtus agrestis*, *Microtus arvalis*, *Myodes glareolus*, *Sorex araneus*, *S. coronatus*, and *S. minutus*) were collected at four high incidence sites (H1-H4) and four low (L1-L4) incidence sites for PUUV infection in humans. Kidney-derived DNA samples were tested for *Leptospira* spp. by real-time PCR targeting the *lipl 32* gene and further analyzed by duplex PCR targeting the *flaB* and the *secY* genes. For the detection of Seewis virus, lung-derived DNA was tested via RT-PCR targeting the nucleocapsid gene.

Results: Altogether, 42 of the 736 small mammals including 27 of 660 bank voles and 11 of 66 shrews, were positive for *Leptospira* spp., while *Sorex* spp. (14.7%) showed significantly higher prevalences compared to bank voles (4.1%). Detected *Leptospira* spp. were pathogenic species other than *L. kirschneri*. Significantly more *Leptospira*-positive bank voles were found at H sites than at L sites. Altogether 22.2% of positive bank voles were infected with PUUV. Double infection of PUUV and *Leptospira* spp. occurrence in bank voles is 1.86 times (OR = 1.86; 95% CI: 0.72–4.73) more likely than infections with each pathogen separately.

Discussion: *Leptospira*- positive bank voles are focally positively associated with PUUV infection in bank voles and with human hantavirus cases. It should be considered that shrews may serve as *Leptospira* spp. reservoirs.

Keywords: Leptospirosis, Puumala, rodent-borne

Introduction

Leptospira spp. are helical-shaped bacteria, which are causative agents for leptospirosis, a zoonotic disease. As 10 pathogenic *Leptospira* species are grouped in 20 serogroups and 300 serovars are known, identical serovars may belong to different *Leptospira* species (Levett 2001). Leptospirosis is considered the most widespread zoonotic disease worldwide (Vijayachari et al. 2008). Earlier, it was described to be a disease of occupational risk for veterinarians, farmers, and rodent control workers in Europe (Levett 2001, Bharti et al. 2003). Today, leptospirosis is linked with increasing recreational outdoor activities, such as water sports (Jansen et al. 2005, Mayer-Scholl et al. 2014) and is therefore of

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emerging concern for humans in temperate regions. Though leptospirosis is known to cause a broad variety of clinical symptoms, its clinical picture has also been described to mimic hemorrhagic fever with renal syndrome (HFRS), which is caused by a group of Old World hantaviruses including the bank vole (*Myodes glareolus*)-associated *Puumala virus* (PUUV) (Lundkvist et al. 1997, Goris et al. 2013, Clement et al. 2014, Mayer-Scholl et al. 2014). PUUV is causing the majority of human hantavirus disease cases in Germany and is endemic in south-western Germany (Ettinger et al. 2012). In addition, *Tula virus* (TULV), another vole-borne hantavirus, and Seewis virus (SWSV), a shrew-borne hantavirus, have been detected in south-western Germany (Schlegel et al. 2012, Schmidt et al. 2016). PUUV and *Leptospira* spp. can cause fever, headache, anuria, conjunctivitis, and myalgia and almost identical kidney value abnormalities in humans (Sion et al. 2002). However, most infections caused by *Leptospira* spp. and PUUV are mis- or underdiagnosed because most symptoms are nonspecific and because of the lacking awareness especially in Europe (Goris et al. 2013, Klempa et al. 2013, Clement et al. 2014, Mayer-Scholl et al. 2014). The incidence of notified leptospirosis cases per year is 0.115/100,000 averaged for the years 2012 and 2013 in Germany. It is 0.105/100,000 also averaged for 2012 and 2013 for the federal state of Baden-Wuerttemberg, south-western Germany (Hoffmeister et al. 2010, Robert Koch Institute: SurvStat@RKI 2.0, https://survstat.rki.de, accessed: June17, 2016), which was examined in the present study. Hantavirus incidences in humans vary greatly spatially and temporally. The average notified incidence for the years 2012/2013 was 1.94/100,000 for Germany and in particular 8.63/100,000 for Baden-Wuerttemberg where the most notified human cases ($n = 5321$ cases in the years between 2001 and 2015) were reported in comparison to other German states (Robert Koch Institute: SurvStat@RKI 2.0, https://survstat.rki.de, accessed: June 17, 2016). Both, *Leptospira* spp. and PUUV are transmitted through direct or indirect contact (excretions of infected small mammals and contaminated water or soil) (Dournon et al. 1984, Yanagihara et al. 1985, Tsai et al. 1987, Kallio et al. 2006, Reis et al. 2008). Small mammals are the most important reservoirs for both, PUUV and several pathogenic *Leptospira* spp. (Faine et al. 1999, Levett 2001, Meerburg et al. 2009, Hughes et al. 2014, Mayer-Scholl et al. 2014). As clinical and epidemiological similarities are known for leptospirosis and PUUV-infection, we conducted this study to determine the prevalence of pathogenic *Leptospira* spp. in small mammals, which have been analyzed for the presence of PUUV before and co-infections of the two pathogens.

The objectives of this study:

(i) Detection and comparison of prevalences of pathogenic *Leptospira* spp. in small mammal species from sites with high and low hantavirus incidences in humans,
(ii) Quantification of hantavirus (PUUV and SWSV)-*Leptospira* spp. co-infections in small mammals.

Materials and Methods

Small mammal samples

In 2012 (between June and November) and 2013 (between March and November), small mammals were collected at four sites with high incidence of notified human hantavirus cases (sites H1-H4) and four sites with low notified incidence (sites L1-L4) in Baden-Wuerttemberg, Germany (Drewes et al. in press; Fig. 1a and Table 1). In total 736 small mammals of seven species were available for the present study (*Myodes glareolus* ($n = 660$), *Sorex coronatus* ($n = 44$), *Sorex araneus* ($n = 22$), *Apodemus flavicollis* ($n = 4$), *Microtus agrestis* ($n = 3$), *Sorex minutus* ($n = 2$), and *Microtus arvalis* ($n = 1$)). In 2012, 526 small mammals (499 *Myodes glareolus*, 13 *S. coronatus*, 7 *S. araneus*, 3 *Microtus agrestis*, 3 *A. flavicollis*, and 1 *Microtus arvalis*) were trapped with a range of 38–122 animals per site. In 2013, 210 small mammals (161 *Myodes glareolus*, 31 *S. coronatus*, 15 *S. araneus*, 2 *S. minutus*, and 1 *A. flavicollis)* were captured. Further information on trapping methods were published before (Drewes et al. in press).

DNA and RNA extraction

For DNA extraction, 600–800 μL phosphate-buffered saline (PBS, pH: 7.2) was added to 0.01–0.02 g of each small mammals' kidney tissue, depending on the initial kidney sample weight. Subsequently, 0.6 g of sterile ceramic beads (PeqLab Biotechnologie GmbH, Erlangen, Germany) (diameter sized 1.4 mm) were added to each sample. Samples were homogenized for 3×15 s at 100 Hz by the Precellys® 24 Tissue Homogenizer (Bertin Technologies, Montigny-le-Bretonneux, France). Afterward, DNA was extracted with the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) as recommended by the manufacturer. Quantity and quality of the extracted DNA samples were determined with a spectrophotometer (Peqlab Biotechnologie GmbH, NanoDrop 2000c).

For RT-PCR detection of SWSV in shrews, RNA was isolated from lung tissue by QIAzol® Lysis Reagent (Qiagen) treatment as described by Schmidt et al. (2016).

Samples were tested for PUUV by PCR and ELISA in a previous study (Drewes et al. in press).

Altogether, 90of 660 *M. glareolus* (13.64%; 95% CI: 11.2–16.5) were previously detected positive for PUUV. In 2012, 85 of 499 *M. glareolus* (17.0%; 95% CI: 14.0–20.6) were positive for PUUV and in 2013, five out of 161 *M. glareolus* (3.1%; 95% CI: 1.1–7.3) were tested positive (Drewes et al. in press).

PCR methods for the detection of Leptospira *spp.*

A real-time PCR targeting the partial *lipl 32* gene for pathogenic *Leptospira* spp. was performed with the primers LipL32-45F (5′-AAG CAT TACCGC TTG TGG TG-3′) and LipL32-286R (5′-GAA CTC CCA TTT CAG CGA TT-3′) following a modified protocol using the Mx3000P QPCR System (Agilent Technologies, Santa Clara, CA) (Stoddard et al. 2009, Woll et al. 2012). We modified the PCR as previously described by Woll et al. (2012) by using a different probe (Univ 18 s-p: 5′ YAK-CGCGGTTCTATTTTGTTG GT BBQ-3′). This PCR method detects following pathogenic *Leptospira* spp.: *L. interrogans*, *L. kirschneri*, *L. weilii*, *L. noguchii*, *L. borgpetersenii*, *L. santarosai*, *L. alstoni*, and *L. alexanderi*. Its limit of detection was 20 genomic equivalents/reaction with a 95% cutoff value. To distinguish *L. kirschneri* from other pathogenic *Leptospira* spp., samples were further investigated by duplex PCR using primers G1 (5′-CTG AAT CGC TGT ATA AAA GT-3′), G2 (5′-GGA AAA CAA ATG GTC GGA AG-3′), B64 I (5′-ACT AAC TGA GAA ACT TCT AC-3′), and B64 II (5′-TCC TTA AGT CGA ACC TAT GA-3′)

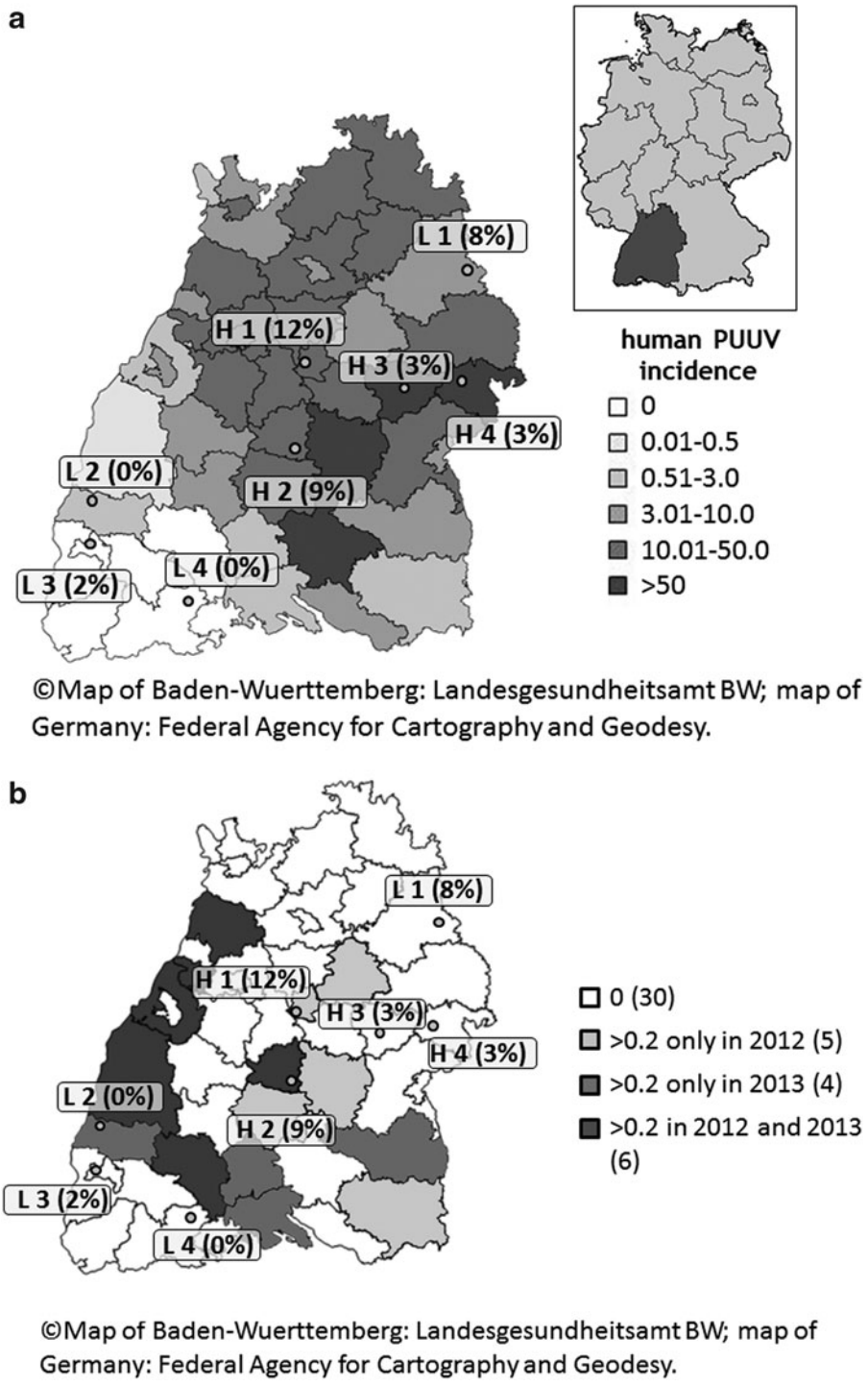


FIG. 1. **(a)** Percentage prevalences of Leptospira spp. in small mammals collected at districts with high incidences (H) and low incidences (L) for human hantavirus infection from Baden-Wuerttemberg, south-west Germany (taken from Drewes et al. submitted, modified) Legend: H1 = Stuttgart; H2 = Tübingen; H3 = Göppingen; H4 = Heidenheim; L1 = Schwäbisch Hall; L2 = Emmendingen; L3 = Freiburg im Breisgau; L4 = Waldshut **(b)** Human leptospirosis incidences per district in Baden-Wuerttemberg, Germany, accessed for the years 2012 and 2013 (according to Robert Koch Institute: SurvStat@RKI 2.0, https://survstat.rki.de, accessed: June 27, 2016 for the years 2012 and 2013) Legend:H1 = Stuttgart; H2 = Tübingen; H3 = Göppingen; H4 = Heidenheim; L1 = Schwäbisch Hall; L2 = Emmendingen; L3 = Freiburg im Breisgau; L4 = Waldshut.

(Gravekamp et al. 1993). The duplex PCR is targeting the partial flagellin-encoding *flaB* gene (563 base pairs, bp), which exclusively amplifies in *L. kirschneri* and the preprotein translocase-encoding *secY* gene (285 bp), which amplifies in several different pathogenic *Leptospira* spp. such as *L. interrogans*, *L. weilii*, *L. noguchii*, *L. borgpetersenii*, *L. santarosai*, and *L. meyeri* (Gravekamp et al. 1993). We used this PCR method to distinguish *L. kirschneri* from other pathogenic *Leptospira* species. Due to low amounts of *Leptospira* DNA detected by real-time PCR, only 18 of 42 samples yielded a result by duplex PCR.

RT-PCR method for the detection of SWSV

Serological and RT-PCR analyses for PUUV and TULV in bank voles and in other rodents have been described previously (Drewes et al. in press). RT-PCR detection of SWSV in shrews and sequencing of amplification products followed a previously described standard protocol with slight modifications (Schlegel et al. 2012) using SuperScript®III One-Step RT-PCR kit with Platinum®Taq (Invitrogen, Carlsbad, CA) and the BigDye® Terminator version 1.1 Cycle Sequencing-Kit (Applied Biosystems, Darmstadt, Germany).

Table 1. Detection of *Leptospira* spp. DNA in Small Mammals Collected at the Eight Trapping Sites with High (H1-H4) and Low (L1-L4) Human Hantavirus Incidences During 2012 and 2013 from Baden-Wuerttemberg, Germany

	Number of positive small mammals/total number of small mammals per study site and year (percentage of small mammals positive for Leptospira *spp.)*																	
	H1		*H2*		*H3*		*H4*		*L1*		*L2*		*L3*		*L4*		*Total*	
Small mammal species	2012	2013	2012	2013	2012	2013	2012	2013	2012	2013	2012	2013	2012	2013	2012	2013		
Bank vole (*Myodes glareolus*)	10/87 (12)	1/12 (8)	4/80 (5)	2/15 (13)	2/80 (2.5)	0/9 (0)	1/68 (1.5)	0/15 (0)	5/55 (9)	1/44 (2)	0/32 (0)	0/6 (0)	2/40 (5)	0/46 (0)	0/57 (0)	0/14 (0)	27/660 (4.1)	
Common shrew (*Sorex araneus*)	1/2 (50)	0/5 (0)	—	0/1 (0)	0/1 (0)	0/2 (0)		0/1 (0)	0/1 (0)	1/4 (25)	—	—	0/1 (0)	—	0/1 (0)	0/2 (0)	3/22 (14)	
Crowned shrew (*Sorex coronatus*)	1/4 (25)	2/10 (20)	1/2 (50)	3/9 (33)	0/5 (0)	1/7 (14)	0/1 (0)	1/3 (33)	0/2 (0)	0/1 (0)	—	—	—	—	—	0/1 (0)	8/44 (18)	
Pygmy shrew (*Sorex minutus*)	—	—	—	0/1 (0)	—	—	—	0/1 (0)	—	—	—	—	—	—	—	—	0/2 (0)	
Voles (*Microtus* spp.[a])	1[b]/11 (100)	—	—	—	0/1 (0)[b]	—	1[b]/1 (100)	—	—	—	—	—	—	—	0/1 (0)	—	2/4 (50)	
Yellow-necked mouse (*Apodemus flavicollis*)	0/1 (0)	—	—	—	—	—	—	—	2/2 (100)	0/1 (0)	—	—	—	—	—	—	2/4 (50)	
Total	16/122 (13.1)		10/108 (9.3)		3/105 (2.9)		3/90 (3.3)		9/110 (8.2)		0/38 (0)		2/87 (2.3)		0/76 (0)		42/736 (5.7)	

[a]Microtus arvalis ($n=1$) and Microtus agrestis ($n=3$).
[b]Microtus arvalis.

For RT-PCR novel S segment-specific primers SWSV_S_ 400_fw 5′-ATC TTG CTY AAR GGA CTC TAC ATG CT and SWSV_S_ 890 rev 5′-TCA GGY ACW GAC TGG ACA GCA GG were used and the annealing temperature was adjusted to 56°C. The obtained SWSV sequence was deposited in GenBank under Acc. No. KX447717.

Statistical analysis

Confidence intervals (95% CI) for prevalences of *Leptospira* and hantavirus in small mammals were determined by the Clopper and Pearson method using the Graph Pad Software (Graph Pad Software, Inc., San Diego, CA). Fisher's exact test was used with a type I error α of 0.05 to test the independence of compared prevalences for *Leptospira* spp. (Graph Pad Software, Inc.). The odds ratio (OR) with 95% CI were computed to measure the association between *Leptospira* spp. infection and PUUV infection in bank voles.

Results

PCR analysis for pathogenic Leptospira *spp.*

In total 42 of 736 small mammal DNA samples were tested positive for *Leptospira* by real-time PCR (5.7%; 95% CI: 4.2–7.6; Table 1). *Leptospira* spp. was found in all small mammal species, except *S. minutus* and *Microtus agrestis*. The average prevalences in 2012 (5.7%; 95% CI: 4.0–8.0) and 2013 (5.7%; 95% CI: 3.2–9.8) were identical ($p=0.13$). Out of 660 bank voles, 27 were positive for *Leptospira* spp. (4.1%; 95% CI: 2.8–5.9). The prevalence in shrews (14.7% 95% CI: 9.4–27.6) was significantly higher than in voles (*Myodes glareolus* and *Microtus* spp.) ($p=0.001$). The prevalence in the 44 crowned shrews was (18.2%; 95% CI: 9.3–32.2) and in the 22 common shrews it was 13.6% (95% CI: 3.9–34.2). The two pygmy shrews were *Leptospira* negative (0%; 95% CI: 0–71.0). Moreover, two out of four yellow-necked mice and likewise two out of four individuals belonging to the genus *Microtus* spp. were positive for *Leptospira* spp.(50%; 95% CI: 15–85). Altogether 18 of 42 samples were further tested by duplex PCR and identified to be pathogenic *Leptospira* spp. other than *L. kirschneri*.

Positive small mammal DNA samples were found at six (sites H1-4 and sites L1, L3) of the examined eight study sites (Fig. 1a, b). The average prevalence was significantly lower at the L sites (2.7%; 95% CI: 1.3–5.4); than at the H sites (6.7%; 95% CI: 4.3–10.3) ($p=0.03$). The prevalences of H and L sites did not differ significantly between the years 2012 and 2013 ($p=0.13$). Prevalences per site and small mammal species are shown in Table 1.

Analysis of Leptospira*–hantavirus co-infections*

Double infections of PUUV and *Leptospira* spp. (22.2%; 95% CI: 10.3–41.1) occurred in six out of 27 Leptospira positive bank voles. Double infection of PUUV and Leptospira spp. occurrence is 1.86 times (OR = 1.86; 95% CI: 0.73–4.7) more likely than separate infections either with Leptospira spp. or PUUV in bank voles. All co-infected bank voles originated from H sites in 2012. Altogether, 50% of *Leptospira*-positive bank voles from site H1 were co-infected with PUUV (95% CI: 23.7–76.3). *Leptospira* co-infections were detected in PUUV-ELISA-positive and in PUUV-RT-PCR-positive bank vole samples. Hantavirus-specific

maternal antibodies did not impact on results as all co-infected bank voles were adults according to their body weight (Drewes et al. in press).

In the present study, one out of 22 *S. araneus* (4.6%; 95% CI: 0–23.5) was detected to be SWSV-RT-PCR positive, whereas the two pygmy shrews (*S. minutus*) (95% CI: 0–71.0) and the 44 crowned shrews (*S. coronatus*) (95% CI: 0–9.6) were negative. The SWSV-infected common shrew was not co-infected with *Leptospira* spp.

Discussion and Conclusions

This study demonstrated an average *Leptospira* prevalence of 5.7% for 736 small mammals, with positive bank voles (4.1%), shrews (14.7%), *Microtus* spp. (50%), and yellow-necked mice (50%). The average prevalence observed here (0–50%) is in a range of prevalences previously obtained from small mammals in Germany (Mayer-Scholl et al. 2014, Heuser et al. 2016, Obiegala et al. 2016). *Leptospira* spp. were previously also detected in a similar wide prevalence range (2.9–71.4%) in small mammal species including *Apodemus* spp., *Microtus arvalis*, *Myodes glareolus*, *Mus musculus*, *Rattus norvegicus*, and *Sorex* spp. from other European countries such as Switzerland, The Netherlands, Croatia, Denmark, and Austria (Hartskeerl et al. 1996, Adler et al. 2002, Turk et al. 2003, Schmidt et al. 2014, Heuser et al. 2016). Varying prevalences were also detected in *Rattus norvegicus*, which were collected in The Netherlands, Austria, Hungary, and Germany where human cases were reported (Roczek et al. 2008, Goris et al. 2013, Heuser et al. 2016). In a previous study from Germany, moderate *Leptospira* spp. prevalences of 12–14% were detected in *Microtus arvalis* and *Apodemus agrarius* (Mayer-Scholl et al. 2014). While a previous study from Germany reported the absence of *Leptospira* spp. in shrews belonging to the family Soricidae (*Crocidura russula* $n=4$; *S. araneus* $n=7$; *S. coronatus* $n=1$) (Obiegala et al. 2016), the prevalence in *Sorex* spp. obtained in the present study was 14.71%. Other studies from Germany and Switzerland reported similar prevalences (11–20%) in shrews of the family Soricidae (Adler et al. 2002, Mayer-Scholl et al. 2014). Though voles (family Cricetidae), mice and rats (family Muridae) are known to be main reservoirs for pathogenic *Leptospira* spp., it should be taken into account that shrews (family Soricidae) may likewise serve as reservoirs.

A former study showed that beech mast promotes population growth of bank voles (*Myodes glareolus*) in the subsequent year and that the resulting high bank vole abundance is positively associated to the number of reported human PUUV cases (Reil et al. 2015). The in-transformed proportional cover of beech and oak forest per district and the categorical variable of beech mast in the previous year (yes/no) were determined by our study group in a former study on hantavirus infection in the same small mammal samples (Drewes et al. in press). This former study demonstrated that the PUUV incidences in humans and the PUUV prevalence in bank voles, which were also investigated in the present study, is positively correlated with a combination of beech mast from the previous year and the proportion of forest cover in the respective district in Baden-Wuerttemberg, Germany (Drewes et al. 2016).

Beech mast in 2011 was the likely cause for high bank vole population densities and associated high PUUV prevalences in 2012 in contrast to the year 2013, when bank vole populations collapsed (no beech mast in 2012) (Drewes et al. 2016).

The prevalences of *Leptospira* spp. detected in all small mammals were nearly the same in 2012 and 2013, which suggests that neither bank vole abundance nor beech mast have influence on the spreading and maintenance of *Leptospira* spp. Beech mast is promoting population growth not only in bank vole populations but also in populations of yellow-necked mice, wood mice, and field voles (Imholt et al. 2016). However, it should be that *Leptospira* spp. are pathogens with a variety of reservoir host species that are not affected by beech mast whereas PUUV is exclusively harbored and transmitted by bank voles. This can explain the multi-annual dynamics of PUUV outbreaks in contrast to the more constant *Leptospira* prevalence in small mammals in forest environments.

In a previous study the PUUV prevalence in bank voles was significantly higher at sites with high hantavirus incidences in humans (Drewes et al. in press). In the present study the prevalences for *Leptospira* spp. in bank voles were also significantly higher at H sites with high human hantavirus incidences compared to L sites with low human hantavirus incidences. The notified incidences for leptospirosis in humans differed between 0.5 and 19 among districts and years (Fig. 1b and Table 2).The incidences for leptospirosis in 2012/2013 were slightly, but not significantly higher in the H districts compared to the L districts (Robert Koch Institute:

Table 2. Notified Human Leptospirosis and Puumala Virus Incidences Per 100,000 Citizens and Per District and Year

	Leptospirosis incidences per district		*PUUV incidences per district*	
District (site)	Year 2012	Year 2013	Year 2012	Year 2013
Stuttgart (H1)	1.0	6.0	19.7	0.0
Tübingen (H2)	8.8	19.0	40.0	0.0
Göppingen (H3)	1.2	2.0	56.1	0.4
Heidenheim (H4)	1.6	6.0	58.8	1.6
Schwäbisch Hall (L1)	0.5	1.0	7.5	1.1
Emmendingen (L2)	1.9	3.0	0.6	0.0
Freiburg (L3)	3.7	8.0	0.5	0.0
Waldshut (L4)	1.2	2.0	0.0	0.0

According to: Robert Koch Institute: SurvStat@RKI 2.0, https://survstat.rki.de, Accessed: June 26, 2016 for the years 2012 and 2013.

SurvStat@RKI 2.0, https://survstat.rki.de, accessed: June26, 2016 for the years 2012 and 2013). Overall, 22.2% of all *Leptospira*-positive bank voles were co-infected with PUUV in the present study. However, the distribution of co-infected bank voles appears heterogeneous as all co-infected bank voles were found at H sites. A much lower prevalence of co-infection was reported in bank voles (3.03%) from an area in Croatia where *Leptospira*- and PUUV infections occur sympatrically (Cvetko et al. 2006).

In earlier studies, the prevalences ranged between 7% and 54.4% for SWSV in *Sorex* spp. from Germany, Slovakia, Hungary, Finland, and the Czech Republic (Kang et al. 2009, Schlegel et al. 2012). The prevalence obtained from *Sorex* spp. in the present study is much lower (1.5%). The SWSV nucleotide sequence from the common shrew showed the highest similarity of 84% to a SWSV sequence originating from a common shrew in Ceske Budejovice, Czech Republic (Schlegel et al. 2012). Moreover, the prevalence was not positively linked to areas with high hantavirus incidences in humans, in line with the assumption that SWSV is not pathogenic to humans.

In conclusion, this study confirms bank voles, yellow-necked mice, and field voles as reservoirs of *Leptospira* spp. in south-western Germany. Moreover, this study points out that shrews may likewise serve as *Leptospira* spp. reservoir hosts. Results from this study suggest that the prevalence of pathogenic *Leptospira* spp. is positively linked to PUUV-infection in bank voles and to related high incidence areas of PUUV-infection in humans. As the prevalences were high and restricted to co-infections with PUUV at the H sites and as leptospirosis is causing a variety of nonspecific clinical symptoms, patients with an appropriate case history, such as previous exposure to contaminated water, should be regarded infected with either *Leptospira* spp. and/or PUUV.

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Author Disclosure Statement

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